

A numerical obstruction to Conjecture 00000007914

ColdSmile

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Abstract

The asserted asymptotic for the increasing finite ranks of a covolume spectrum is impossible. We prove a stronger statement for every set of nonnegative real numbers, allowing zero and every real leading constant. The contradiction concerns the numerical assertions themselves and does not use any classification of arithmetic Fuchsian or triangle groups. The accompanying Lean 4 project checks the exact rank argument, the logarithmic asymptotic, and the nonnegativity interfaces for volumes and limits.

1 Original assertion and scope

The English original defines S_{arith} as the “limit-from-below spectrum” of covolumes $\text{vol}(\Gamma \backslash H)$ of arithmetic Fuchsian groups. It asserts that S_{arith} is well ordered (no descending chains), and that its k -th element satisfies

$$v_k \sim c k^{-1} (\log k)^{-2/3} \quad (k \rightarrow \infty). \quad (1)$$

It additionally asserts that the smallest noncompact arithmetic triangle group is $(2, 3, 7)$ and makes a next-smallest trichotomy claim with covolume ratio $3/7$. Both language versions contain the same displayed asymptotic and ordered-spectrum assertions. The exact bilingual source is included as `conjecture.md`.

This report disproves the conjunction by refuting its finite-rank asymptotic. It makes no separate claim about the triangle groups, the trichotomy, or well-ordering in isolation. Throughout, an element’s rank means its rank in the inherited increasing numerical order, as in the original well-order assertion. The formula is treated as an asymptotic for the elements themselves, not for a reciprocal or a counting function.

2 Universal obstruction

For $S \subseteq \mathbb{R}$, say that x has finite rank n if $x \in S$ and the *entire* predecessor set

$$P_S(x) = \{y \in S : y < x\}$$

is finite and has cardinality n . The first element has rank zero. This definition neither counts groups with repeated equal covolumes nor requires all of S to have finite rank.

Lemma 1. *If v_k has rank $k-1$ in S for each integer $k \geq 1$, then $v_m < v_n$ whenever $1 \leq m < n$.*

Proof. If $v_n \leq v_m$, then $P_S(v_n) \subseteq P_S(v_m)$. Their finite cardinalities would give $n-1 \leq m-1$, a contradiction. $\square$

For a fixed $c \in \mathbb{R}$, put

$$g_c(k) = c k^{-1} (\log k)^{-2/3} \quad (k \geq 2).$$

The logarithm is natural and the power is the usual real power of the positive number $\log k$.

Lemma 2. *For every $c \in \mathbb{R}$, $g_c(k) \rightarrow 0$ as $k \rightarrow \infty$. If $c \neq 0$ and $v_k/g_c(k) \rightarrow 1$, then $v_k \rightarrow 0$.*

Proof. The reciprocal k^{-1} tends to zero. Also $\log k$ tends to positive infinity, so $(\log k)^{-2/3}$ tends to zero. The product, multiplied by the fixed constant c , therefore tends to zero. In the second assertion, write $v_k = (v_k/g_c(k))g_c(k)$ for $k \geq 2$ and take limits. $\square$

Theorem 1. *Let $S \subseteq [0, \infty)$. There is no sequence whose k -th term has rank $k - 1$ in S for every $k \geq 1$ and which satisfies (1) in the ordinary ratio-limit sense, for any nonzero real constant c .*

Proof. Lemma 1 gives $0 \leq v_1 < v_2$ and $v_k \geq v_2 > 0$ for every $k \geq 2$. Lemma 2 gives $v_k \rightarrow 0$. Passing to the limit in the eventual inequality $v_2 \leq v_k$ gives $v_2 \leq 0$, contradicting $v_2 > 0$. $\square$

Zero is allowed in S : it is strict separation of the first two finite ranks that supplies a positive lower bound. No monotonicity of the comparison function is transferred through asymptotic equivalence; only its limit is transferred. No exhaustion of a transfinite well-order by a sequence is assumed. The asserted asymptotic presupposes arbitrarily large finite ranks, which is all that is relevant here.

3 Why the obstruction applies to covolumes

A covolume is a nonnegative measure value. Finite covolumes can therefore be regarded as nonnegative real numbers without any geometric classification or positive lower-bound theorem. More explicitly, let (X_i, μ_i) be *any* family of measured spaces and put

$$C = \{\mu_i(X_i) : \mu_i(X_i) < \infty\} \subseteq \mathbb{R}.$$

Then $C \subseteq [0, \infty)$. The Lean interface uses extended-nonnegative real measures and excludes the infinite value before conversion to a real number. In particular, it never turns an infinite covolume into a spurious real zero.

The source does not specify the operation called “limit-from-below spectrum.” We do not choose one favorable construction. The result holds for every nonnegative real set, and in particular for every $S \subseteq \text{cl}(C)$: the closed half-line $[0, \infty)$ contains C and therefore its closure. This covers raw covolumes, their closure, one-sided accumulation values, and arbitrary selections of such values. Strict positivity of a raw covolume is *not* promoted to strict positivity of a limit.

The Chinese wording can also be read as a lower-limit (*liminf*) spectrum. A finite *liminf* of a sequence of nonnegative covolumes remains nonnegative: every tail has nonnegative infimum, so the supremum of those tail infima is nonnegative. Thus the same obstruction applies to that reading as well. Infinite spectrum values cannot serve as terms of the stated real asymptotic. No interpretation that retains covolumes or their ordinary finite limits can introduce negative real values.

Consequently the actual arithmetic covolumes and the ordinary limiting readings of the source all satisfy the necessary nonnegative condition. Theorem 1 rules out the displayed rank asymptotic for any such spectrum. Conjoining further well-ordering or triangle-group assertions cannot repair the contradiction.

4 Formal statement and verification interface

The project pins Lean 4.19.0 and Mathlib v4.19.0 at revision

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The main file is `lean/CovolumeSpectrum.lean`.

- `HasRank` uses a finite set whose membership is exactly membership in the full predecessor set, and whose cardinality is the specified rank. `rank_strictMono` proves Lemma 1.
- `scale` is the literal displayed reciprocal-logarithmic expression with exponent $-(2/3)$ in $\mathbb{R}$. `scale_tendsto_zero` proves its limit.
- `asymptotic_iff_ratio` identifies Mathlib’s asymptotic equivalence with the ordinary ratio limit whenever $c \neq 0$. The totalized values at indices zero and one are irrelevant to the filter at infinity.
- `no_rankedAsymptotic` proves the universal obstruction. It even allows every $c \in \mathbb{R}$ under Mathlib’s extended definition $v - g = o(g)$. For $c = 0$, this requires eventual equality to zero and is also impossible; the ordinary ratio definition itself excludes $c = 0$.
- `no_numericalConjecture` also includes the original well-foundedness requirement. `no_full_conjunction` permits an arbitrary proposition for all remaining source clauses.
- `finiteMeasureValues_nonnegative`
`no_measure_spectrum_conjunction`
These lemmas supply the general measure-volume interface and the closure-of-volume-values conclusion. The geometric quotient construction is not assumed as an additional theorem: the universal obstruction applies to any family of measures.
- `finite_liminf_nonnegative` works with the extended-real liminf and an explicitly finite real value. `no_liminf_spectrum_conjunction` handles selections of finite sequential lower limits directly, without imposing closure as an extra assumption or requiring a separate boundedness hypothesis.

The source-to-formal correspondence uses only the ordinary meanings of volume, finite numerical rank, and asymptotic equivalence. The unnecessary arithmetic classifications are not formalized and are not claimed as verified. The exact commands, theorem-dependency output, independent review, and final validation records accompany the package. These are local verification results; maintainer acceptance is separate.